

E-Commerce US Dataset Analysis

Report 1: Dataset Understanding

Nithish | AI/ML Engineer

1. Project Overview

- Dataset: US e-commerce transactional data representing a mid-scale B2C marketplace.
- Full dataset: 1,000,000 orders; analysis uses a 10,000-order relational sample extracted for fast processing.
- Coverage: 10 US metro areas across 10 states.
- 8 relational tables covering orders, customers, products, sellers, payments, reviews, and geolocation.

2. Table Inventory

Table	Full Rows	Sample Rows	Columns	Primary Key
orders	1,000,000	10,000	8	order_id
customers	1,000,000	10,000	9	customer_id
order_items	2,200,000	21,800	8	order_id + order_item_id
order_payments	1,150,000	11,500	5	order_id + payment_sequential
order_reviews	933,700	9,300	7	review_id
products	2,000	1,980	10	product_id
sellers	500	500	8	seller_id
geolocation	11,500	11,200	5	zip_code_prefix

3. Table-by-Table Description

3.1 Orders

- Tracks lifecycle of each order from purchase to delivery.
- Key columns: order_id, customer_id, order_status, order_purchase_timestamp, order_approved_at, order_estimated_delivery_date, order_delivered_carrier_date, order_delivered_customer_date.
- Only two statuses present: delivered (93.3%) and canceled (6.7%).
- All date columns stored as strings in the source — require conversion.

3.2 Customers

- Contains demographic and geographic profile for each buyer.
- Key columns: customer_id, customer_unique_id, customer_name, customer_gender, customer_age, customer_zip_code_prefix, customer_city, customer_state, customer_segment.
- 10,000 customer_id values but 7,898 unique customer_unique_id values — 21% of customers placed more than one order.
- Three customer segments: Consumer (69.8%), Corporate (20.5%), VIP (9.8%).

3.3 Order Items

- Line-item detail — one row per product per order.
- Key columns: order_id, order_item_id, product_id, seller_id, price, freight_value, discount_rate, shipping_limit_date.
- Average 2.2 items per order; maximum 6 items in a single order.
- Zero missing values across all 21,800 sample rows.

3.4 Order Payments

- Records each payment transaction; one order may have multiple rows (split payments).
- Key columns: order_id, payment_sequential, payment_type, payment_installments, payment_value.
- 7 payment methods: credit card (50.3%), PayPal (13%), bank transfer (9.8%), debit card (8.9%), Apple Pay (8.6%), voucher (6.8%), boleto (2.5%).

- 15% of orders span more than one payment row — require aggregation before joining.

3.5 Order Reviews

- Post-delivery feedback submitted by customers.
- Key columns: review_id, order_id, review_score, review_comment_title, review_comment_message, review_creation_date, review_answer_timestamp.
- 93.4% review coverage; 668 missing reviews correspond exactly to canceled orders.
- review_comment_title is categorical (14 fixed labels), not free text.

3.6 Products

- Product catalog — 2,000 SKUs across 7 categories and 8 brands.
- Key columns: product_id, product_category_name, product_brand, product_name, cost, price, product_weight_g, product_length_cm, product_height_cm, product_width_cm.
- Both cost and price are present, enabling direct margin calculation.
- All 4 dimension columns fully populated with no nulls.

3.7 Sellers

- 500 sellers; distributed across the same 10 cities as customers.
- Key columns: seller_id, seller_company_name, seller_contact_name, seller_contact_gender, seller_contact_age, seller_zip_code_prefix, seller_city, seller_state.
- 7 sellers share parent companies — multi-brand operators.

3.8 Geolocation

- Maps zip code prefixes to latitude, longitude, city, and state.
- 11,500 unique zip codes; zero nulls — cleanest table in the dataset.
- Coordinate precision (6 decimal places) is sufficient for distance calculations.

4. Table Relationships

From Table	To Table	Join Key	Cardinality
orders	customers	customer_id	Many-to-One
orders	order_items	order_id	One-to-Many
orders	order_payments	order_id	One-to-Many
orders	order_reviews	order_id	One-to-Many
order_items	products	product_id	Many-to-One
order_items	sellers	seller_id	Many-to-One
customers	geolocation	zip_code_prefix	Many-to-One
sellers	geolocation	seller_zip_code_prefix	Many-to-One

5. Data Dictionary

- A 60-row metadata file (data_dictionary.csv) documents every column across all 8 tables.
- Fields: table_name, column_name, data_type, description, example_value, nullable, primary/foreign key flags.
- Serves as the single reference for column semantics throughout the project.

6. Geographic Coverage

State	City	Orders (%)
California	Los Angeles / San Francisco	25.5%

Texas	Houston / Dallas	18.9%
Florida	Miami	13.1%
New York	New York City	~10%
Illinois	Chicago	~8%
Pennsylvania	Philadelphia	~6%
Georgia	Atlanta	~5%
Colorado	Denver	~3%
Washington	Seattle	~3%
Arizona	Phoenix	~2%